

Ideal Sampling

$x(t)$ is a low pass signal (a smooth function)
 W is its bandwidth (all frequencies describing $x(t)$)

Ideal sampling is performed by multiplying $x(t)$ with a Dirac delta train

$$x_s(t) = x(t) \times \sum_{k=-\infty}^{+\infty} \delta(t - kT_s)$$

where T_s is a sampling interval.

Illustration

Signal as a function of time:

$$x(t) = \cos(2\pi t) + 2\cos(6\pi t)$$

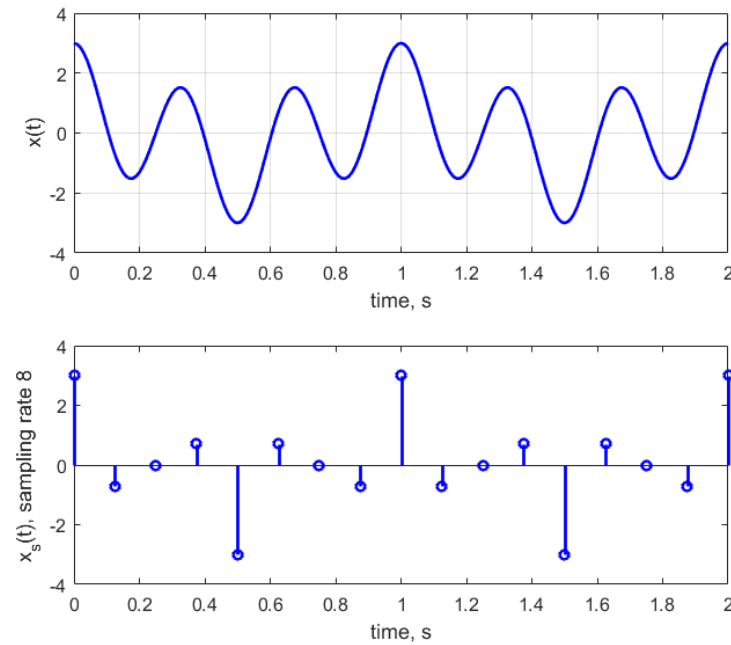
Its Fourier transform:

$$X(f) = \frac{1}{2}\delta(f \pm 1) + \delta(f \pm 3)$$

If we treat it as a low pass signal (described by low frequencies), then its bandwidth is $W = 3$.

Let us sample the signal at the rate $f_s = 8$ Hz (the sampling period is $T_s = 1/8$ s).

Illustration



FT of sampled signal

FT of the sampled signal:

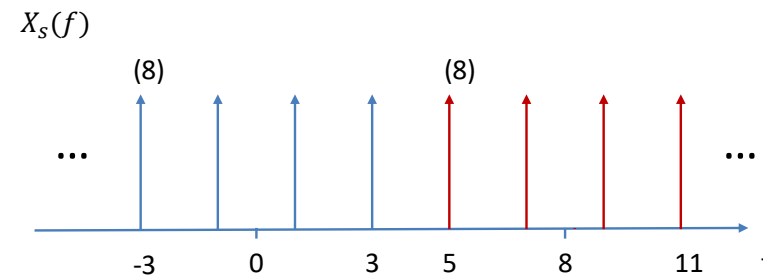
$$X_s(f) = X(f) * \sum_{k=-\infty}^{+\infty} f_s \delta(f - kf_s)$$

“*” stands for correlation and $f_s = 1/T_s$
It can also be written as:

$$X_s(f) = \sum_{k=-\infty}^{+\infty} f_s X(f - kf_s)$$

Many copies of $X(f)$!

Back to illustration



FT of the sampled signal

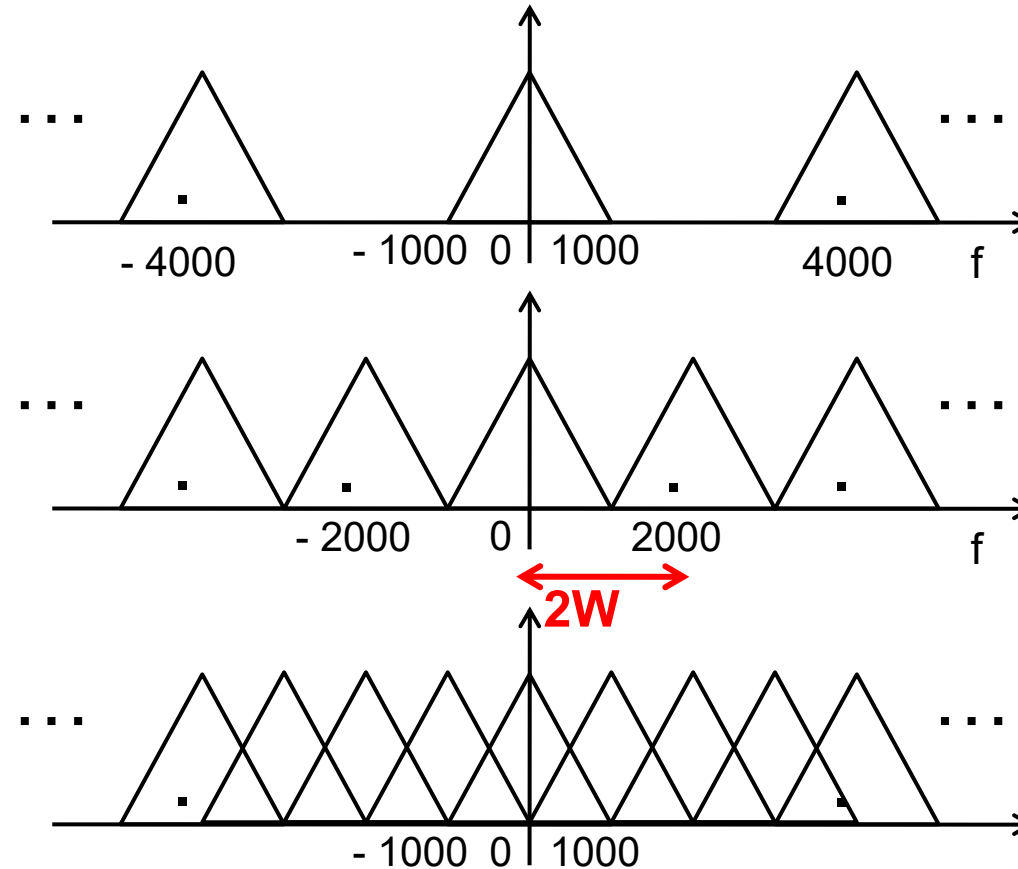
$$x(t) = \cos(2\pi t) + 2\cos(6\pi t)$$

sampled at the rate $f_s = 8$ Hz is

$$X_s(f) = \sum_{k=-\infty}^{+\infty} (4\delta(f \pm 1 - k8) + 8\delta(f \pm 3 - k8))$$

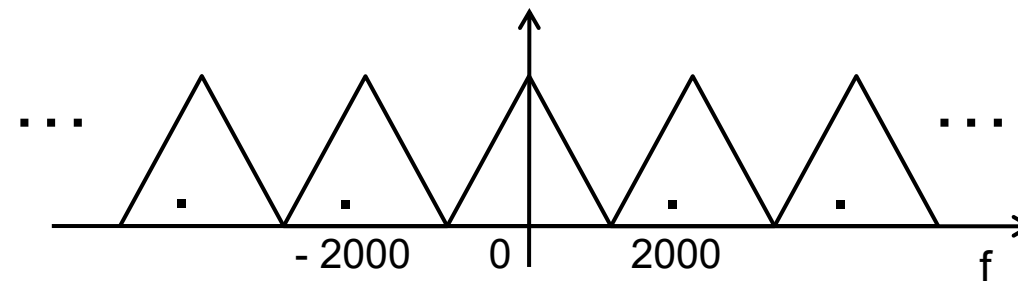
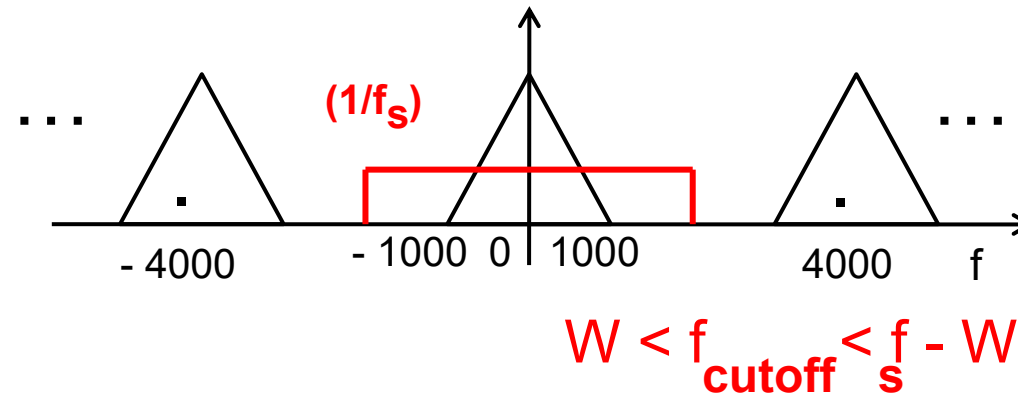
Nyquist Sampling Theorem

$W=1000$ is the bandwidth of a band pass signal
Find the smallest value of sampling rate such that spectral copies do not overlap.

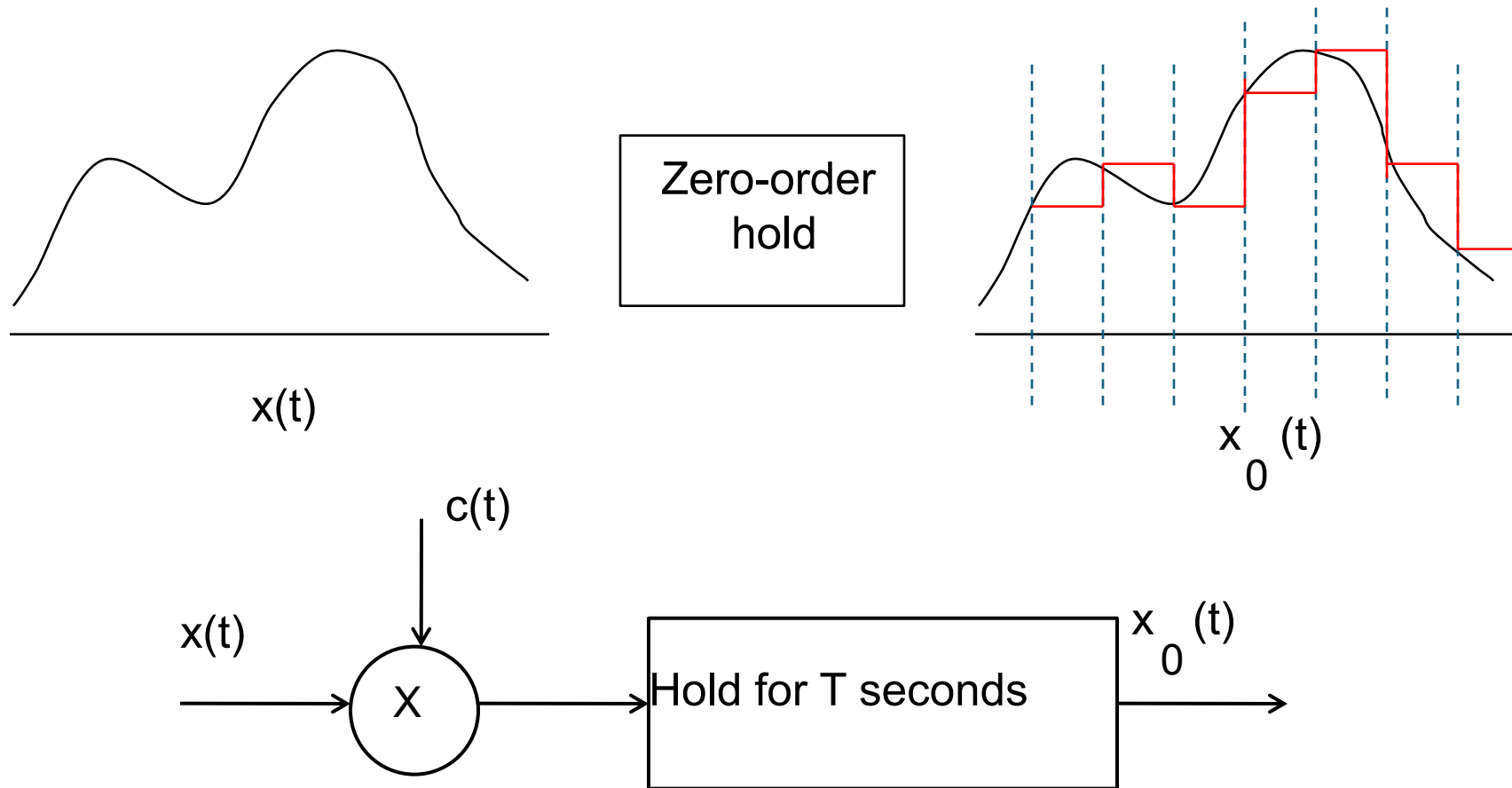


Nyquist Sampling Theorem

Assume that sampling frequency is high (spectral copies do not overlap)
How to recover the original spectrum?



Sampling with Zero-Order Hold



Aliasing

- Left panel: Properly Sampled image of brick wall (493-by-599)
- Right panel: Spatial aliasing in the form of a [Moiré pattern](#) (205-by-250)
- <http://en.wikipedia.org/wiki/Aliasing>



Aliasing (another example)



Aliasing due to undersampling

Application of anti-aliasing filter

